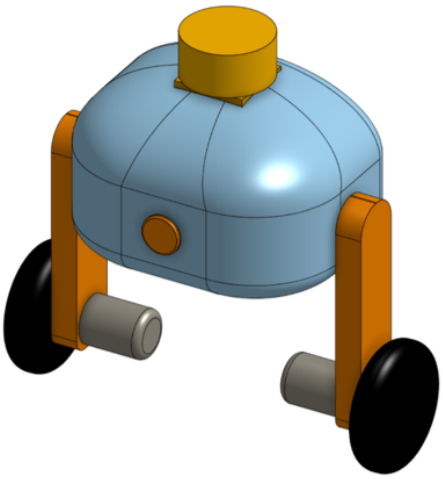
Gyrobot use cases report



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1 Introduction

As part of our fourth year at Polytech Orléans engineering school, we are required to undertake one of the projects provided or to propose one of our own. We therefore proposed to build a robot that balances on two wheels using a gyroscope: the Gyrobot.

It is a compact, self-balancing, two-wheeled robot, designed to operate in environments where mobility, accessibility, and autonomous navigation can pose challenges.

The project aims to explore the design of a small, lightweight robotic platform capable of maintaining its balance on two wheels, whilst serving as a basis for future applications such as mapping, exploration, and assistance in challenging environments.

In this report, we will cover the general description of the robot, the user experience, the algorithmic operation, the use case, and finally the Gantt chart.

2 General description of the robot

The robot can be divided into three main parts:

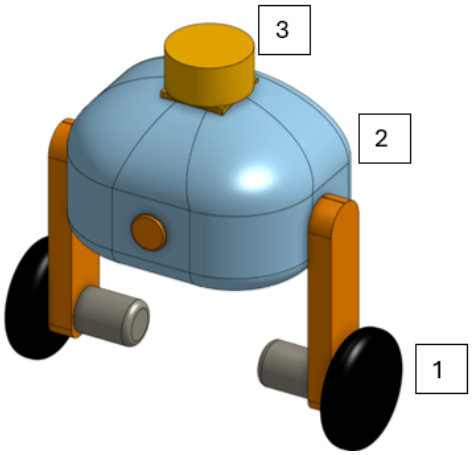


Figure 1: 3D sketch of the Gyrobot

The lower part of the robot, the motors, wheels and 'legs' marked **1** in the image below. This part must be robust so it can withstand the rigours of various types of terrain, as well as the impacts caused by the robot's movement.

The central section of the robot, labelled **2**, contains the various components required for movement, balance, and processing of lidar data. It also houses the battery needed to power the robot. The biggest challenge with this section will be managing the weight and its distribution.

The upper section of the robot, designated **3**, will house a tool required for mapping. This section must include at least two different tools in order to make the robot more versatile.

The project also includes a user interface to display the data transmitted and processed by the robot.

3 User experience

This section focuses on what the user needs to do to use our product.

3.1 Essential feature of the map

The Gyrobot will need to move around, map its surroundings, process the data and send it back to a user interface. It will also need to receive the commands required to move the robot. That is why we have opted to use two cards, to avoid a potential data overload if there were only one card.

A map will need to manage commands relating to the robot's movements. It must therefore:

- To be able to transmit the commands required for movement wirelessly
- Transmit data such as speed, the robot's angle relative to the ground and remaining battery charge
- Heat resistance
- To be fairly small in size
- To be energy-efficient

The second card will be responsible for processing mapping-related data and sending it to our HMI. It will therefore need to:

- To have sufficient computing power to process lidar data
- To be able to communicate wirelessly with the HMI
- Heat resistance
- To be fairly small in size
- To be energy-efficient

There are various boards to choose from: an Arduino, an ESP 32, a Raspberry Pi, an STM 32 and a Tensy.

The selection of the various components is carried out in: Description of technical choices

3.2 HMI

We also need to develop a user interface to display the data returned by the robot. The image below was created partly with the help of generative AI.

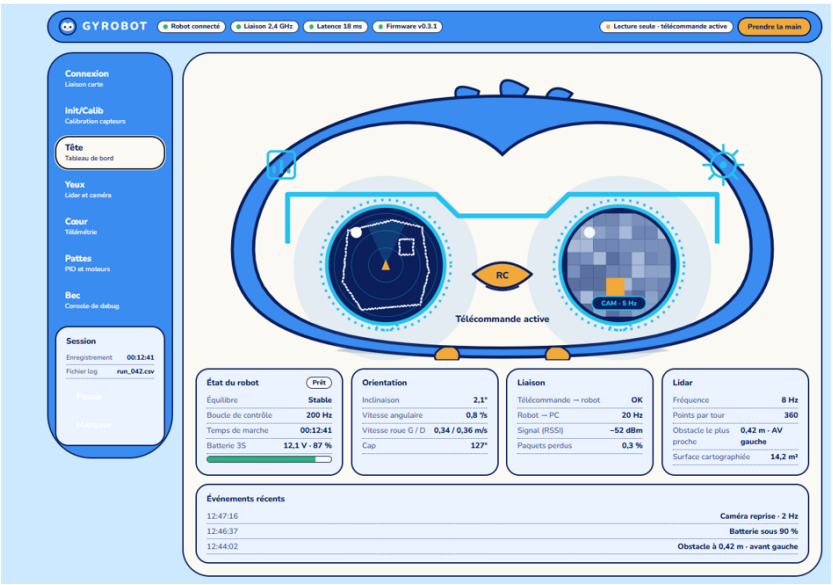


Figure 2: Image from the IMH

The user interface will comprise several sections/tabs to monitor the various values and lidar data returned by the robot. This penguin-themed user interface was chosen by our team and is purely for aesthetic purposes.

The layout and the data displayed in the user interface are subject to change.

4 Algorithmic operation

4.1 Flow chart

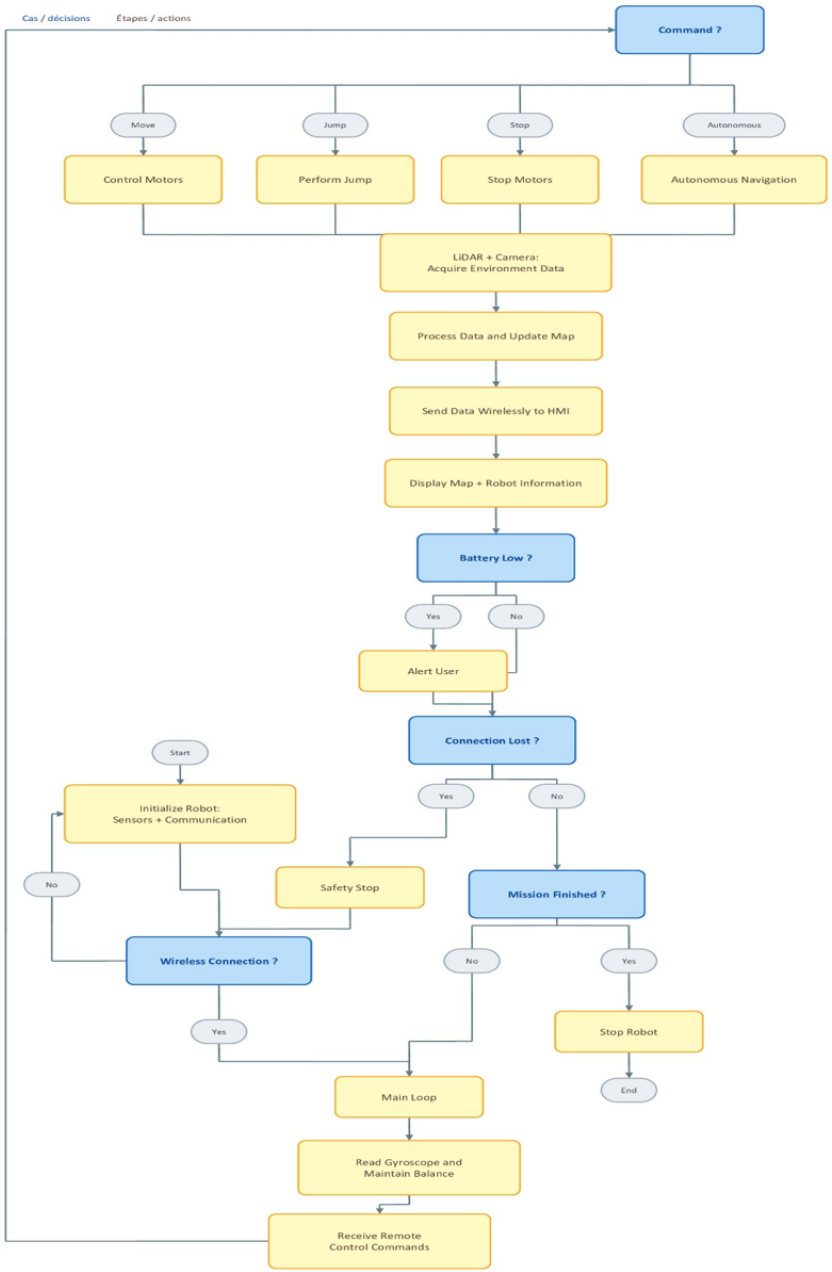


Figure 3: Flowchart of the Gyrobot

This flowchart explains how the programme's algorithm works.

5 Specifications

In order to draw up the specifications correctly, we need to identify the project's requirements and expectations. As regards expectations, we need to build a robot that can balance on two wheels, is capable of mapping its environment, and can interact via a remote control and a user interface that displays the robot's data. As for the requirements, we have defined two types: primary requirements, which are necessary for the robot's basic operation, and secondary requirements, which serve to make the robot more versatile.

The primary requirements align with our expectations, as we have set them ourselves.

As for the secondary requirements, we wish to develop a user interface and a robot resembling a penguin, to have a rechargeable battery, and for the lidar to be replaceable with a high-resolution camera.

6 Retro planning (Gantt)

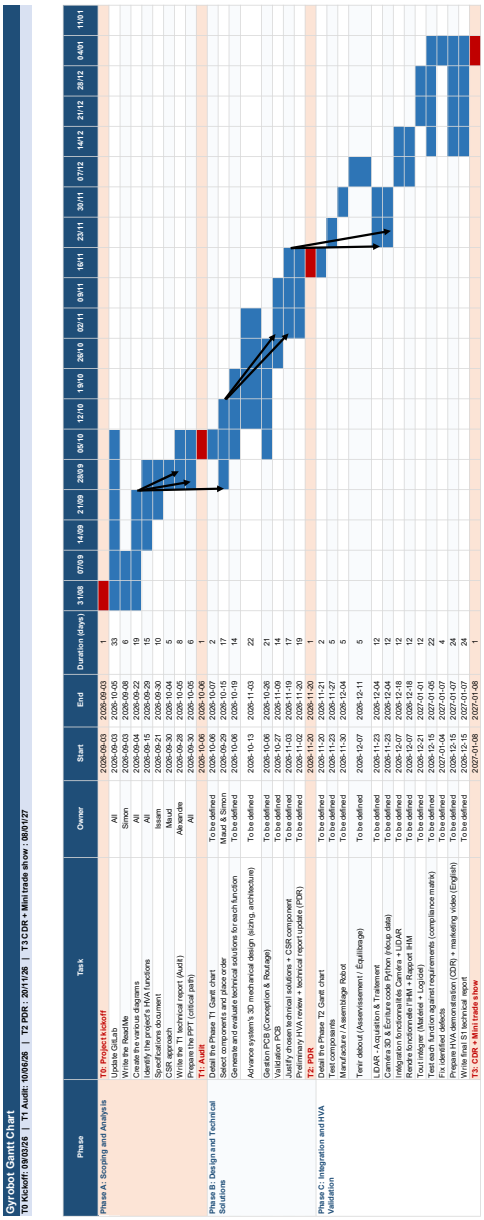


Figure 4: Gantt chart (updated on 21 September 2026)

This schedule will be updated as we make progress on the project. In terms of organisation, our biggest challenge is that we only have three months to build a prototype.
This means we need to have a very precise and rigorous organisational structure.

7 Validation testing and functional analysis

7.1 Functional analysis

7.1.1 Horned beast diagram

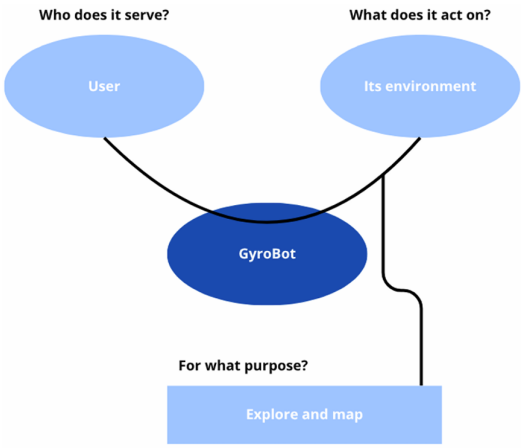


Figure 5: Horned beast diagram

The horned beast diagram provides a clear understanding of a user's needs before designing a product or service. In our case, our aim is to map an area and create a map of it, all using the Gyrobot. It will be controlled by a user in various environments, both indoors and outdoors.

7.1.2 Octopus diagram

Here are the various block diagrams showing the critical and main functions.

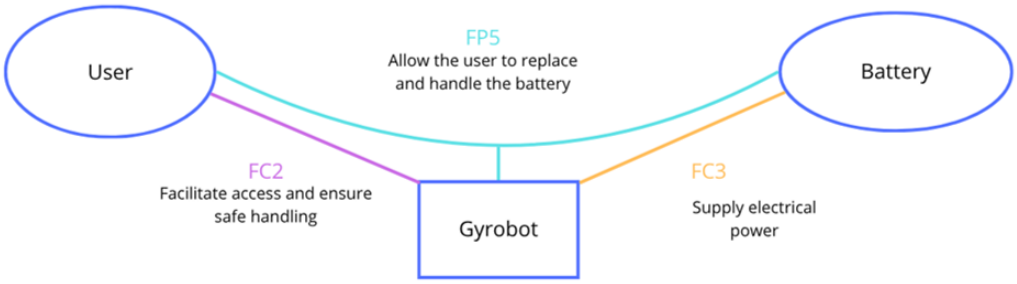


Figure 6: Energy recharge phase

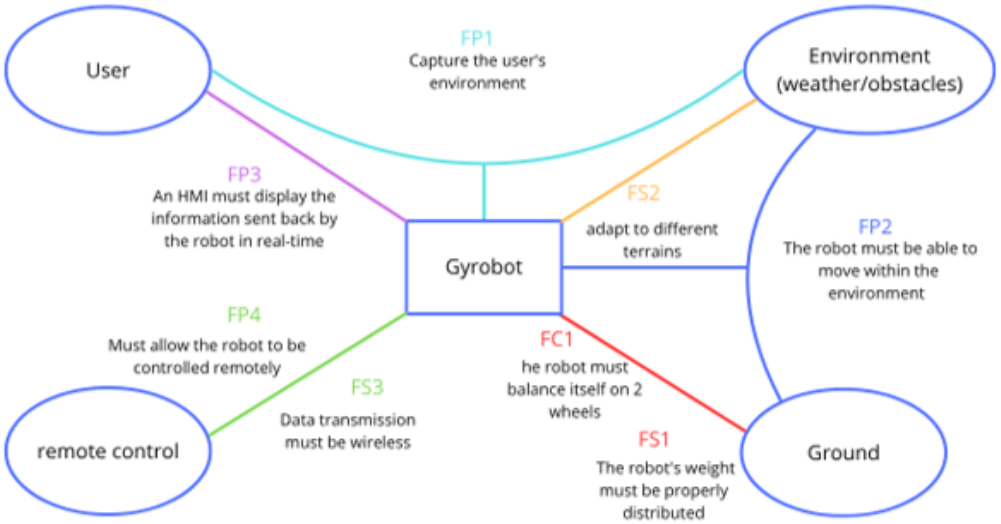


Figure 7: Usage phase

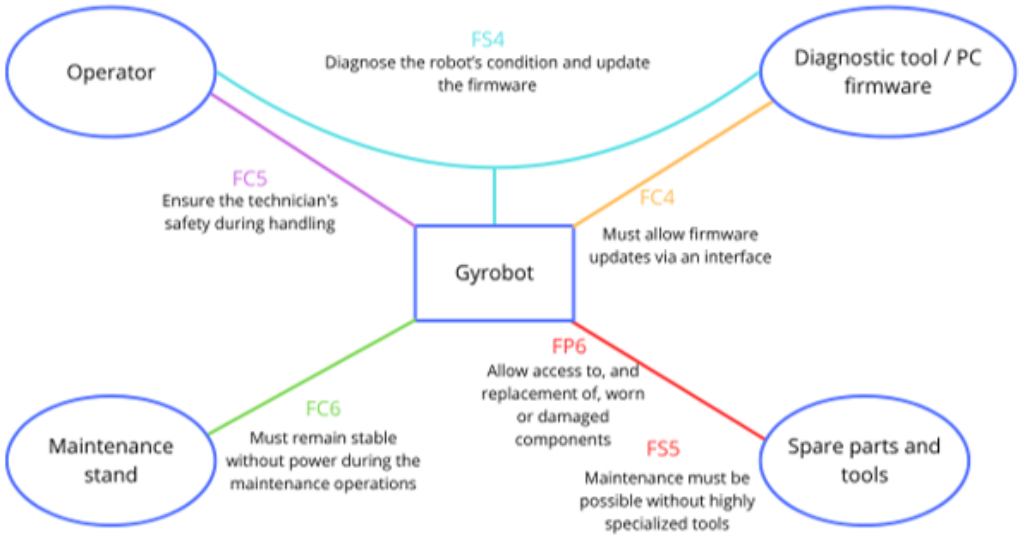


Figure 8: Maintenance phase

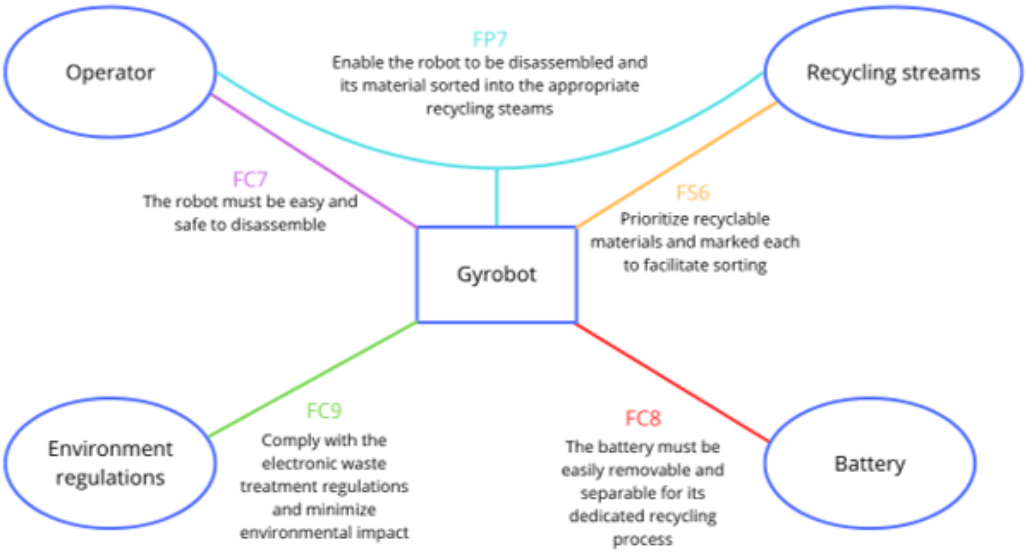


Figure 9: Recycling phase

7.1.3 FAST diagrams

Using octopus diagrams, we obtain the FAST diagram.



Figure 10: Fast diagrams

The FAST diagram enables us to work our way from our requirements to the technical solutions.

7.2 Functional table

Using the FAST diagram, we can draw up a functional diagram of the Gyrobot.

Criteria	Level	Tolerance
Map accuracy/ precision	5 cm	-3cm + 5cm
LiDAR max range	10m	± 2m
Slip	None	/
Tip-over / Fall	None	/
Speed	4 km/h	± 2 km/h
Minimum turning radius (at 4 km/h)	0.5 m	± 0.2m
Display latency	< 100 ms	/
Map refresh rate	2 Hz	± 1 Hz
Radio control range	> 30 m (interior)	/
	> 150 m (exterior)	/
Command response time	< 50 ms	/
Replacement time	< 1 min	/
Replacement time per component	< 1 hour	/
Tools required	Standard tools	/
Full disassembly time	< 1 hour	
Number of separable material families	< 5	+ 2
Maximum climbable slope	> 30°	/
Tools required for access	Standard tools	/
Connector safety	?	/
Minimum autonomy	> 3min	/
Nominal voltage	?	/
Interface type	USB+Wifi	/
Supported file format	?	/
Power cut-off before handling	Manual	/
Sharp edge	None	/
Presence of a dedicated stand	Obligatory	/
Stand holding position	Horizontal	/
Reclinable	45°	± 15°
Stand load capacity	> 1,5 kg	/
Tools required for disassembly	Standard tools	/
Removal time	< 2 min	/
Tool-free removal	Yes	/
Regulatory compliance	?	/
Hazardous substances presence	None	/
Center of Gravity (CoG) position	Centered, above the wheel axis	/
Maximum total weight	< 1,5 kg	/
Maximum obstacle height	20 cm	± 5cm
Success rate of landing without falling	80%	± 10 %
Minimum bandwidth from the robot to remote control	> 30 kbps	/
Minimum bandwidth from the robot to the HMI	> 2 Mbps	/
Frequency	Wi-Fi 2.4 GHz	/
Anomaly detection coverage	80%	/
Firmware update duraction	< 5 min	/
Debug communication mode	USB+Wifi	/
Tools required for access	Standard tools	/
recyclable material by mass	>50 %	/
Differents marks on each material	Colored	/

Figure 11: Functional analysis

7.3 Validations test

To avoid building the robot from scratch only to find that it doesn't work without knowing why, we need to put validation tests in place to test each major component of the robot separately.

Here are the various tests that have been put in place:

- Connection and acknowledgement test
- Command response time test
- Remote control range test
- Communication reliability test
- Data rate and range test to the computer
- Lidar range test
- Map accuracy test
- Display latency and map refresh rate test
- Balance test
- Speed and turning radius test
- Weight test
- Battery replacement test
- Simultaneous operation test

7.3.1 Connection and acknowledgement test

We send commands from the remote control and check that the ESP32 receives them and sends back an acknowledgement. The test is successful if the green dot remains lit on the remote control.

7.3.2 Testing the response time of the control system

The remote control records the time a command is sent and the time the acknowledgement is received. The difference gives the response time, which

must be less than 50 ms. The statistics are displayed in a ‘Statistics’ tab on the remote control, and the mean and standard deviation are monitored.

7.3.3 Remote control range test

Gradually move the remote control away from the robot, first indoors and then outdoors, with and without obstacles. Record the distance once the connection is lost. The test is successful if the connection remains stable for more than 30 metres indoors and more than 150 metres outdoors. Check the green dot and the number of lost packets in the 'Stats' tab on the remote control.

7.3.4 Transaction reliability test

Leave the robot to run for 1 minute, whilst moving it around, and note the number of lost packets in the 'Statistics' tab on the remote control, then in the HMI. The test is successful if the loss rate remains below 5 per cent.

7.3.5 Testing the data rate and range to the computer

Measure the actual data transfer rate between the Raspberry Pi and the computer. The data transfer rate must remain above 2 Mbps.

7.3.6 Lidar range test

An obstacle is placed at increasing distances from the robot, and we check whether it appears on the HMI map. The maximum range should be 10 m ± 2 m. We compare the distance displayed on the HMI with the measured distance to verify reliability at each distance.

7.3.7 Map accuracy test

We have a room or object mapped, having first measured its dimensions. We then compare the dimensions shown on the map with those in reality. The test is deemed successful if the deviation falls within the tolerance specified in the specifications (5 cm, -3 / +5 cm).

7.3.8 Testing the display latency and refresh rate of the graphics card

The Raspberry Pi stamps each data point with a timestamp upon acquisition, and the computer compares this timestamp with the display time, after synchronising the two. The latency must be less than 100 ms. To calculate the refresh rate, we count the number of map updates over 10 seconds, which should give 2 Hz $\pm$ 1.

7.3.9 Balance test

The robot is left stationary and then set in motion for several minutes, whilst being given gentle nudges. The robot must neither fall over nor slip. We also check that it can climb up and down a slope steeper than 30°.

7.3.10 Speed and turning circle test

A distance of 10 m is marked out and the time taken to cover this distance at maximum speed is recorded: 4 km/h corresponds to approximately 9 s, with a margin of error of $\pm$ 2 km/h. To test the cornering, the robot is made to trace a circle, and its radius is measured (0.5 m $\pm$ 0.2 m). A visual check is carried out to ensure there is no slippage.

7.3.11 Weight test

We weigh the complete robot, including the battery, on a set of scales: it must weigh less than 1.5 kg.

7.3.12 Battery replacement test

Someone who did not design the robot replaces the battery, timing the process, without using any tools. The replacement must take less than a minute. A check is then carried out to ensure the robot restarts correctly.

7.3.13 Simultaneous testing

We run everything at the same time: balance, movement, controls, Lidar, camera and data transmission to the computer. We monitor the stats tabs for several minutes. The test is successful if the robot does not fall over and if all the previous criteria are met.

8 Description of technical choices

8.1 Remote control

Remote control – Needs analysis (FP4)
FP4 : the remote control must be able to pilot the robot remotely

1. Criteria of function FP4 (from the specifications)

Criterion	Level	Unit	Source
Minimum radio range (indoor)	30	m	Specifications : function table (FP4)
Minimum radio range (outdoor)	150	m	Specifications : function table (FP4)
Maximum response time of the control	50	ms	Specifications : function table (FP4)

2. Input/output needs of the remote control (based on the electronic diagram)

Control element	No. of elements	Analog. channels / element	Digital inputs / element	Digital outputs / element	I2C pins / element	Total analog	Total digital inputs	Total digital outputs	Total I2C pins	Role on the robot (to be completed)
2-axis joystick (KY-023)	2	2	1	0	0	4	2	0	0	Joystick 1 : forward/backward (Y axis) + turn right/left (X axis) ; Joystick 2 : tilt right/left (X axis) + lower/raise (Y axis)
Slide potentiometer	3	1	0	0	0	3	0	0	0	Adjustment of the robot's 3 PID controller gains (P, I, D)
Rotary potentiometer	1	1	0	0	0	1	0	0	0	Adjustment of the maximum travel speed
Push button	5	0	1	0	0	0	5	0	0	1 button = jump ; 4 buttons = macros / screen navigation
Signal LED	1	0	0	1	0	0	0	1	0	Power-on status
OLED screen (I2C)	1	0	0	0	2	0	0	0	2	Menu navigation, communication parameter settings, robot information display (speed, battery, ...)
On / off switch	1	0	0	0	0	0	0	0	0	In series with the power supply
Digital voltmeter	1	0	0	0	0	0	0	0	0	Connected to the battery
TOTAL						**8**	**7**	**1**	**2**	

3. Summary : what the remote control must provide (microcontroller selection criteria)

Indicator	Value	Comment
Analog channels to acquire	8	2 joysticks × 2 axes + 3 sliders + 1 rotary
Digital inputs (buttons)	7	5 buttons + 2 buttons integrated into the joysticks
Digital outputs	1	Signal LED
I2C pins (SDA + SCL)	2	OLED screen
Total GPIO pins required	**18**	The microcontroller must provide at least this number of usable GPIOs
Values to be transmitted per radio frame	15	Analog channels + button states (to be sent to the robot)
Minimum transmission frequency (Hz)	20	At least 1 complete transmission per maximum response time
Selected transmission frequency (Hz)	50	Assumption : 50 Hz (×2.5 margin over the minimum)
Transmission period (ms)	20,0	Maximum delay between two frames
Estimated radio latency + robot processing (ms)	10	Assumption : ESP-NOW ≈ a few ms + robot-side reading
Estimated response time (ms)	30,0	Transmission period + latency

Figure 12: Remote control

Remote control - Component selection

Justification of each component with respect to function FP4 (piloting the robot remotely)

No.	Function performed	Component selected	Reference / model	Main characteristics	Justification of choice	Alternatives considered	Alternative drawbacks	Qty	Unit price (€)	Total price (€)
1	**Command processing (acquisition, computation, screen interface)**	Microcontroller	ESP32-S3-DevKitC-1-N16R8	Dual-core Xtensa LX7 240 MHz · 16 MB Flash + 8 MB PSRAM, 3.3 V logic supply, many GPIOs including 10 channels usable on ADC1	Enough GPIOs and ADC1 channels for the 18 pins required (see sheet « Requirements »).	Arduino Nano	No built-in radio, little memory, fewer ADC inputs	1	8,00 €	8,00 €
						ESP32-S3 version N8R2/N8R8	Less Flash/PSRAM, lower memory margin			
						STM32	More processing power but radio and Wi-Fi must be added as an external module			
2	**Radio link control → robot**	Communication protocol	ESP-NOW (2.4 GHz Wi-Fi, native on the ESP32-S3)	Typical latency of a few ms, potential range with this board : > 200 m outdoor with no obstacles, > 100 m outdoor with obstacles, > 20 m indoor	Latency compatible with the required response time (< 50 ms). Built into the ESP32-S3 : no external radio module to wire.	Detailed comparison against other solutions in the sheet « Radio Comparison ».				
3	**Proportional control of movement and orientation**	2-axis joystick	Module KY-023	2 10 kΩ potentiometers (X and Y axes) + built-in push button, 3.3-5 V supply	Proportional analog signal. Joystick 1 = forward/backward (Y axis) and turn right/left (X axis) ; Joystick 2 = tilt right/left (X axis) and lower/raise (Y axis).	Directional buttons	On/off only, no gradual control	2	4,00 €	8,00 €
4	**Proportional adjustment of the controller gains**	Slide potentiometer	Linear slide potentiometer	10 kΩ linear resistive track, 3 pins	PID position readable at a glance, gradual and precise adjustment, 1 analog channel each.	Rotary potentiometer	Less readable	3	1,60 €	4,80 €
						Incremental encoder	More connectors used per value to change			
5	**Speed adjustment**	Rotary potentiometer	VRB-101M10	10 kΩ, 3 pins	Gradual and compact adjustment, easy to read for speed	Slide potentiometer	Bulkier and less readable	1	0,50 €	0,50 €
						Rotary encoder	No physical stop, more connectors used			
6	**On/off commands**	Push button	BP-GT1-ON	Momentary open contact, read with the ESP32's internal pull-	Simple, reliable, low-cost, 1 digital input	Toggle switch	Does not natively return to its initial position	5	0,30 €	1,50 €

Figure 13: Remote control – Component selection

Remote control – Comparison of radio / microcontroller solutions
Choice of the wireless link solution (FP4 elimination criteria, then weighted scoring)

A. Solutions considered with typical orders of magnitude and compliance with FP4

Criterion	Unit	ESP32-S3 + ESP-NOW	Arduino Nano + nRF24L01 (PA/LNA)	Arduino + LoRa (SX1278)	Bluetooth module HC-05
Frequency band		2.4 GHz	2.4 GHz	433 / 868 MHz	2.4 GHz
Typical outdoor range (direct line of sight)	m	200	800	2000	10
Typical link latency	ms	5	3	200	30
Meets the FP4 outdoor range ?		YES	YES	YES	NO
Meets the FP4 response time ?		YES	YES	NO	YES
Compliant with function FP4 ?		**YES**	**YES**	**NO**	**NO**

B. Weighted scoring (0 = very unfavorable … 5 = very favorable) – indicative scores to be adjusted

Criterion	Weight	ESP32-S3 + ESP-NOW	Arduino Nano + nRF24L01 (PA/LNA)	Arduino + LoRa (SX1278)	Bluetooth module HC-05
Outdoor range	25%	4	5	5	1
Latency / responsiveness	20%	5	5	1	3
Number of I/Os and analog channels	15%	5	3	3	3
Ease of implementation (wiring, code)	15%	4	3	2	4
Cost	10%	5	4	2	4
Power consumption	10%	3	4	4	3
Integration / size	5%	5	3	2	3
Total weight (must equal 100 %)	**100%**				
Score out of 20		17,6	16,4	11,6	11,0
Score retained (0 if not compliant with FP4)		17,6	16,4	0,0	0,0
Rank		1	2	3	3
Decision		**SELECTED**	**Not selected**	**Eliminated (FP4)**	**Eliminated (FP4)**

Figure 14: Remote control – Comparison of radio/microcontroller solutions

Remote control – Energy balance and battery life

Justifies the choice of the LiPo battery and the LM2596 converter

1. Power supply parameters

Parameter	Value	Comment
Nominal battery voltage (V)	7,4	Battery selected : 2S LiPo (7.4 V nominal)
Battery capacity (mAh)	500	Battery selected : 2S 500 mAh LiPo
Usable capacity fraction	80%	Assumption : 80 % (to avoid fully discharging the battery)
LM2596 output voltage (V)	5,0	Set to 5 V to power the ESP32 board
Converter efficiency	80%	Assumption : 85 % (typical 80–90 % for an LM2596)

2. Element power consumption

Element	Qty	Unit current (mA)	Powered by	Total current (mA)	Assumption / source
ESP32-S3 (ESP-NOW transmission)	1	120	5 V rail	120,0	Average during Wi-Fi transmission -> higher peaks
OLED screen 0.96" SSD1306	1	20	5 V rail	20,0	Screen on and changing image at 2Hz
Joystick KY-023	2	1	5 V rail	2,0	2 10 kΩ potentiometers at 3.3 V ≈ 0.7 mA
Slide potentiometer	3	0,5	5 V rail	1,5	10 kΩ at 3.3 V ≈ 0.33 mA
Rotary potentiometer	1	0,5	5 V rail	0,5	10 kΩ at 3.3 V ≈ 0.33 mA
Signal LED	1	10	5 V rail	10,0	LED permanently on
Digital voltmeter	1	10	5V rail	10,0	Connected to the output of the 7.4V->5V converter

3. Results

Indicator	Value	Comment
Total current on the 5 V rail (mA)	164,0	Sum of the elements powered by the converter
Power on the 5 V rail (mW)	820	P = U * I
Current drawn from the battery (mA)	138,5	Rail power referred back to the battery (efficiency included)
Estimated battery life (h)	**2,9**	Usable capacity / current drawn
Max. LM2596 current (mA)	3 000	Manufacturer value : 3 A
Voltage margin Vbattery − Voutput (V)	2,4	The LM2596 requires a minimum input difference of about 2 V
Sufficient voltage margin ?	YES	To be checked when the battery is discharged

Figure 15: Remote control – Energy balance and battery life

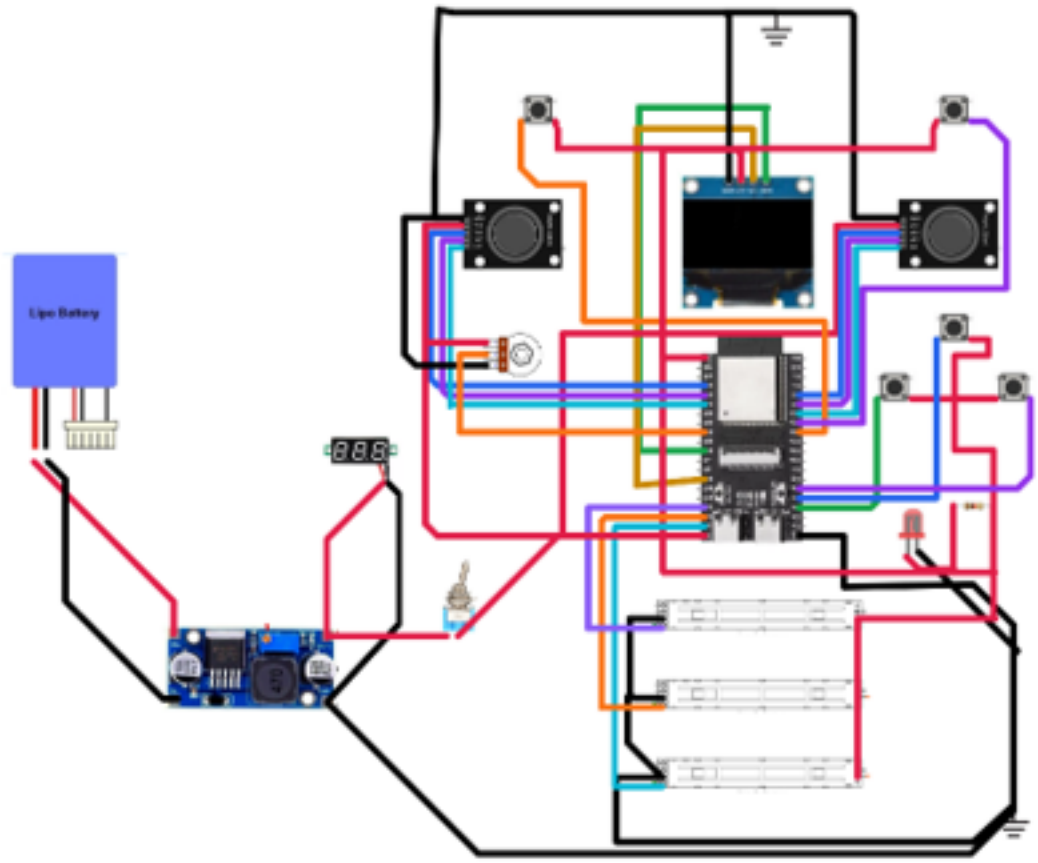


Figure 16: Electronic schematic for the remote

9 Gyrobot

9.1 Motor selection

1. Data and formulas

Data

- Maximum robot mass: $M = 1.3$ kg
- Gravitational acceleration: $g = 9.81$ m/s^2
- Number of motors: 2 identical motors
- Specifications requirement: Nominal speed $v \in [2.0; 6.0]$ km/h (4 ± 2 km/h)
- Fall prevention criterion: Minimum acceleration $a_{\max} > 9.81$ m/s^2 ($1.0\ g$)

Formulas

1. Angular speed: $\omega = N \times (2\pi/60)$ (rad/s)
2. Max linear speed: $v_{\max} = \omega \times R$ (m/s) $= 3.6 \times (\omega \times R)$ (km/h)
3. Total ground thrust (2 motors): $F = \frac{2 \times \tau_{\text{stall}}}{R}$ (N)
4. Max acceleration: $a_{\max} = \frac{F}{M}$ (m/s^2) and Margin in $g = \frac{a_{\max}}{9.81}$

Characteristics of the 3 motors

- **Motor 1 (FIT0520):** $N = 300$ RPM ($\omega = 31.42$ rad/s), $\tau = 3.6$ kg $\cdot$ cm $= 0.353$ N $\cdot$ m ($2\tau = 0.706$ N $\cdot$ m)
- **Motor 2 (FIT0521):** $N = 210$ RPM ($\omega = 21.99$ rad/s), $\tau = 10.0$ kg $\cdot$ cm $= 0.981$ N $\cdot$ m ($2\tau = 1.962$ N $\cdot$ m)
- **Motor 3 (FIT0522):** $N = 100$ RPM ($\omega = 10.47$ rad/s), $\tau = 6.5$ kg $\cdot$ cm $= 0.637$ N $\cdot$ m ($2\tau = 1.275$ N $\cdot$ m)

2. Maximum speed calculations ($v_{\max}$)

Speed conclusion:

Motor	R = 3.5 cm	R = 4.0 cm	R = 5.0 cm	R = 6.0 cm
FIT0520 (300 RPM)	3.96 km/h	4.52 km/h	5.66 km/h	6.79 km/h (*Too fast*)
FIT0521 (210 RPM)	2.77 km/h	3.17 km/h	3.96 km/h	4.75 km/h
FIT0522 (100 RPM)	1.32 km/h	1.51 km/h	1.88 km/h	2.26 km/h

Table 1: Maximum linear speed as a function of wheel radius (allowed range: 2 to 6 km/h).

- **FIT0522 (100 RPM):** Eliminated. Far too slow for wheel radii from 3.5 to 5 cm (< 2.0 km/h, non-compliant with specifications).
- **FIT0520 (300 RPM):** Compliant up to 5 cm, but exceeds the maximum speed limit at 6 cm (6.79 km/h > 6.0 km/h).
- **FIT0521 (210 RPM):** Compliant. Strictly within the required $[2.0 - 6.0]$ km/h range for all evaluated radius.

3. Maximum Acceleration Calculations ($a_{\max}$ for $M = 1.3$ kg)

Total thrust is $F = (2 \times \tau)/R$ *and acceleration is* $a_{\max} = F/1.3$

Motor	R = 3.5 cm	R = 4.0 cm	R = 5.0 cm	R = 6.0 cm
FIT0520 ($3.6\ \mathrm{kg \cdot cm}$)	$15.5\ \mathrm{m/s^2}$ (1.58 g)	$13.6\ \mathrm{m/s^2}$ (1.38 g)	$10.9\ \mathrm{m/s^2}$ (1.11 g)	$9.0\ \mathrm{m/s^2}$ (0.92 g) (*Fall*)
FIT0521 ($10.0\ \mathrm{kg \cdot cm}$)	$43.1\ \mathrm{m/s^2}$ (4.40 g)	$37.7\ \mathrm{m/s^2}$ (3.85 g)	$30.2\ \mathrm{m/s^2}$ (3.08 g)	$25.2\ \mathrm{m/s^2}$ (2.56 g)
FIT0522 ($6.5\ \mathrm{kg \cdot cm}$)	$28.0\ \mathrm{m/s^2}$ (2.86 g)	$24.5\ \mathrm{m/s^2}$ (2.50 g)	$19.6\ \mathrm{m/s^2}$ (2.00 g)	$16.4\ \mathrm{m/s^2}$ (1.67 g)

Table 2: Maximum available acceleration to prevent tipping for a 1.3 kg robot.

Acceleration conclusion:

- To stabilize the inverted pendulum, the acceleration must strictly exceed gravity ($a > 1.0\ g = 9.81\ \mathrm{m/s^2}$).
- **FIT0520 ($3.6\ \mathrm{kg \cdot cm}$):** As soon as the radius reaches 6 cm, acceleration drops to $0.92\ g$ ($< 1.0\ g$): the robot cannot recover and tips over. Even at 4 cm ($1.38\ g$), the margin is very narrow.

- **FIT0521 (10.0 kg · cm):** Delivers between 2.56 g and 4.40 g of acceleration, guaranteeing that the robot will never stall or tip over.

4. Selected Choice

1. **FIT0522 (100 RPM; 6.5 kg · cm):** Rejected because it is far too slow.
2. **FIT0520 (300 RPM; 3.6 kg·cm):** Rejected due to insufficient torque and risk of balancing failure at 1.3 kg.
3. **FIT0521 (210 RPM; 10.0 kg · cm): Selected.** The only motor that simultaneously satisfies the speed requirement and provides ample acceleration margin across all radii.
4. **Optimal Wheel Radius:** $R = 3.5$ cm to 4.5 cm (diameter of 70 to 90 mm), providing ≈ 3.2 km/h and nearly 4 g of acceleration.

9.1.1 electronic schematic

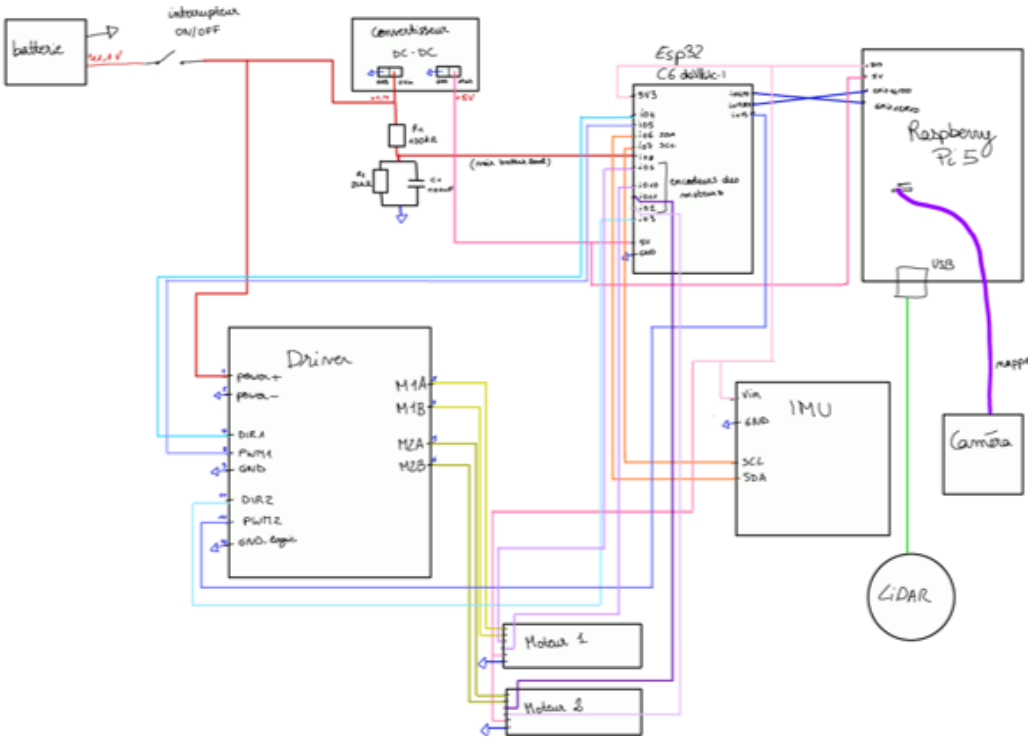


Figure 17: electronic schematic for the robot

9.1.2 Justification of the component's choices

Open the full Excel file, as it is not possible to insert it neatly into the report. (Gyrobot_Component_Selection.xlsx)

10 RSE

The Gyrobot project integrates an eco-design approach to limit our carbon footprint. Instead of making parts far away, all the structure is manufactured locally by 3D printing using PLA, a bio-based and recyclable plastic. To make the sorting easier at the end of life (FS6), recycling symbols are modelled directly in relief on the CAD parts. To go further in accessibility, each material or sorting instruction has its own geometric logo in relief and a colour code. These colours were specifically chosen to be easily seen by colourblind people (yellow, dark blue, white, black, magenta).

For the electronic part, designing and routing a custom motherboard allows us to connect the microcontrollers and components directly through copper tracks. This avoids using dozens of flying wires. It reduces the number of plastic cables used and makes the system much tougher.

The robot was also designed to last in time and to fight against obsolescence (meaning avoiding throwing it away at the first broken component). All the architecture allows us to easily access fragile parts to change them quickly (FP6). The assembly is done using only standard screws, so the structure can be disassembled very simply (FC7) with basic screwdrivers without special tools (FS5). The battery, for example, can be removed very easily to be sent to its specific recycling channel which makes recycling much easier (FC8).

This design choice allows us to respect the European WEEE directive (Waste Electrical and Electronic Equipment) which imposes strict rules to limit the pollution of this kind of system (FC9).

Finally, beyond the environmental impact, the Gyrobot has a real social utility focused on safety at work. The main goal is to map inaccessible, unstable or dangerous places instead of humans. Whether it is to explore a building for a rescue or to inspect unreachable industrial environments, all the necessary spatial data is collected by the robot without ever putting the lives of rescue teams or technicians in danger.

11 FMEA Table

FMEA is an analysis table that helps us figure out what could go wrong when our robot is exploring. Since the robot only uses two wheels, keeping its balance and staying connected to the remote are extremely important. This table lists the main problems the system might face. We give each risk a score based on how bad it is (Severity), how often it happens (Frequency) and how hard it is to spot (Detection). This will help us to focus on the biggest dangers so we can figure out the best ways to prevent them.

Table 3: Failure Mode and Effects Analysis (FMEA) for Gyrobot

Subsystem	Failure Mode	Potential Effect	Potential Cause	S	F	D	RPN	Corrective Action
Balance Control	IMU thermal drift or vibration noise	Loss of absolute tilt reference, robot falls	Motor vibrations, temperature changes	8	3	2	**48**	Low-noise Adafruit LSM6DSOX IMU, software low-pass filter & watchdog
Balance Control	PID loop execution latency	High-frequency oscillation, balance loss	OS scheduler jitter or CPU overload	9	4	2	**72**	Dedicated ESP32-C6 RTOS task (200 Hz hard real-time loop)
Propulsion	Motor stall or torque saturation	Inability to recover tilt, robot tips over	Steep slope, low battery voltage	9	3	3	**81**	High-torque FIT0521 motors (10 kg · cm stall), current monitoring
Propulsion	Wheel slip or traction loss	Odometry drift, balance instability	Low-friction or smooth surface	6	4	3	**72**	High-grip TPU 3D-printed tires, wheel speed anomaly filtering
Communication	ESP-NOW wireless link drop	Loss of remote guidance, runaway risk	Radio interference, out of range	8	3	2	**48**	Automatic failsafe: motor shutdown after 100 ms without signal

Table 3: Failure Mode and Effects Analysis (FMEA) for Gyrobot

Subsystem	Failure Mode	Potential Effect	Potential Cause	S	F	D	RPN	Corrective Action
Navigation	LiDAR optical obstruction	Collision, incorrect 2D SLAM map	Dust on lens, direct sunlight glare	6	3	3	**54**	Camera redundancy, proximity safety thresholds
Power Supply	Sudden LiPo voltage drop	Microcontroller reboot, sudden fall	Deep discharge, cell degradation	10	2	2	**40**	Hardware cutoff threshold (9.9 V), real-time telemetry alert

Legend:

- G (Severity): 1 to 4
- F (Frequency): 1 (Rare) to 4 (Frequent)
- D (non-detection): 1 (Immediately detectable) to 4 (Hard to predict)
- C (Criticality) = G × F × D

12 Conclusion

To conclude, this project is going to be very demanding, particularly in terms of time. Nevertheless, our team has shown great commitment, which is spurring us on to persevere and achieve our goal of producing a prototype before the end of the term, featuring the following key functions:

- To balance,
- Mapping using lidar,
- Move around using a remote control.

13 Appendices

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